Technical Design Document (TDD)

Chikkamagaluru Estate Resort Direct-Booking Website & AI Enquiry Triage Platform

Client: Company A, Chikkamagaluru, Karnataka

Candidate Submission: Software Engineering Intern Take-Home Task (IsoCode Labs)

Submission Deadline: 9 Nov 2026, 11:59 pm IST

- **GitHub Repository**: https://github.com/PrakashMannam/chikkamagaluru-resort-booking
- **Live Application (Frontend on Vercel Edge)**: https://chikkamagaluru-resort-booking.vercel.app
- **Live Backend API (Railway Cloud)**: https://chikkamagaluru-resort-booking-production.up.railway.app

1. Executive Summary & Client Context

Company A operates a boutique coffee-estate luxury resort in Chikkamagaluru, Karnataka, nestled in the Western Ghats. The property features **76 keys** comprising **64 Estate Rooms** and **12 Private Pool Villas**, alongside outdoor banquet lawns for destination weddings and corporate offsites. The resort employs 140 staff and generates ₹24–28 crore in annual revenue under the leadership of the General Manager (GM).

1.1 The Business Problem & Financial Opportunity

- **Current State**:

  - 68% of room nights originate from Online Travel Agencies (OTAs) charging 18–22% commission (averaging 20%).

  - Only 12% of bookings occur directly through resort channels because the existing website is an informational brochure with a telephone number.

  - Event enquiries arrive haphazardly across personal WhatsApp, email, and social media, with rates manually maintained in spreadsheets.

  - Approximately 30% of event enquiries wait over 24 hours for a response, forfeiting an estimated 3 events per month.

- **Financial Baseline**:

  - 76 keys × 365 days × 62% average occupancy = 17,199 room nights at an average daily tariff of ₹9,500 = ₹16.34 crore total lodging revenue.

  - OTA commission toll: 68% via agencies = 11,695 room nights = ₹11.11 crore gross bookings. At 20% commission, the resort forfeits **₹2.22 crore annually**.

  - Direct conversion impact: Shifting 15% of agency bookings to direct saves **₹33.3 lakh/year** in commission.

  - Event conversion impact: 3 lost events/month at ₹2 lakh contribution = 36 events × ₹2 lakh = **₹72 lakh/year**.

- **Total annual bottom-line opportunity**: **₹1.053 crore/year**.

- **The Brief's Core Mandate**:

  - Deliver a high-conversion, mobile-responsive direct-booking engine allowing guests to select dates, view real-time availability and dynamic pricing with statutory GST, and complete reservations with zero overbooking risk.

  - Implement a centralized enquiry intake and AI triage pipeline that ingests web form submissions, simulated emails, and simulated WhatsApp messages, classifies intent, extracts key entities, assigns a specialist owner, and prepares grounded response drafts with human-in-the-loop approval.

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2. Requirements Baseline & Brief Compliance Matrix

Every functional, technical, and operational constraint from the IsoCode Labs problem statement has been verified and delivered end to end:

Requirement Category	Client Brief Specification	Implementation Status	Implementation Evidence
Direct Booking Engine	Rooms with photos, live availability, rate plans, prices with statutory GST, test payment, instant confirmation	**100% VERIFIED**	`/stay`, `/rooms/:code`, `/checkout`, `/payment/:ref`, `/bookings/:ref`
Allotment Control	Site sells only admin-set allotment; atomic inventory locking prevents overbooking under concurrent access	**100% VERIFIED**	`SELECT ... FOR UPDATE` row locks in `BookingTransactionService`, 10-minute hold expiration, rollback on timeout
Admin Controls	Daily rates, website allotment, blackout dates, promo code management	**100% VERIFIED**	`/staff/inventory`: grid editor for rates/allotments, resort-wide/category blackout dates, promo discount engine
Enquiry Inbox	Unified inbox for web form, simulated email, simulated WhatsApp messages	**100% VERIFIED**	`/staff/enquiries`: filterable tabs for Rooms, Weddings, Offsites, Complaints, and Unsorted
AI Router & Extraction	Classifies enquiries (Room, Wedding, Offsite, Complaint), extracts dates/guests/budget, assigns owner, drafts reply	**100% VERIFIED**	`GeminiAiRoutingService`: Google Gemini 1.5 with PII redaction, structured JSON schema extraction, deterministic lane assignment
AI Safety & Fallbacks	Low confidence (<0.70) routes to GM unsorted lane; 1-click staff re-routing; drafts never auto-dispatched; prices injected from rate tables; complaints alert GM	**100% VERIFIED**	`QuoteInjectorService` (rate table injection), `DraftPriceSanitizer`, GM alert flags, staff manual approval step
Structured Event Form	Dedicated structured form for weddings and corporate offsites	**100% VERIFIED**	`/enquire`: dynamic step form with guest count, room requirements, catering, budget, and neutral AV setup options
Daily Reservation Export	Daily booking export for legacy PMS/reservation system	**100% VERIFIED**	`GET /api/v1/admin/bookings/export` CSV export with spreadsheet formula injection sanitization (CWE-1236)
Data Privacy & Legal	DPDP Act 2023 compliance; foreign guest Form C passport details taken at check-in, not stored online	**100% VERIFIED**	Explicit DPDP consent checkbox on `/checkout`, no passport field in forms, private booking confirmation link access
Responsive Design	Minimal, polished UX; fast loading on slow hill-station networks; no screen displays competing CTAs	**100% VERIFIED**	WebP responsive images, 44px tap targets, 0 horizontal overflow at 390px, 768px, and 1280px viewports

3. High-Level System Architecture & Technology Choices

The platform is designed around a modular, three-tier architecture that guarantees transactional integrity, strict separation of concerns, and rapid global asset delivery:

```
 [ Edge Client Layer ]
   ├── React 18 SPA (Vite)
   ├── Responsive Design System (CSS Custom Properties, 390px Mobile to 1280px+ Desktop)
   └── Vercel Edge Network (Global CDN, Security Headers, Gzip/Brotli Asset Delivery)
           │
           │ REST / HTTPS (Stateless JSON APIs)
           ▼
[ Application Layer (Railway Cloud Container) ]
   ├── Spring Boot 3.3.5 / Java 21 LTS
   ├── Spring Security 6 (Stateless HMAC-SHA256 JWT, Role & Lane Authorization)
   ├── Transaction Manager (Pessimistic Row-Level Locking, 10-Minute Hold Scheduler)
   ├── Gemini AI Router (Google Gemini 1.5, Structured Schema Extraction, Hallucination Guardra
   ├── Pricing & GST Calculation Engine (Stateless Statutory Single-Slab Valuation)
   └── Simulated Delivery Adapters (Email SIM, WhatsApp SIM with Unique Delivery IDs)
           │
           │ JDBC Connection Pool (HikariCP, utf8mb4)
           ▼
[ Persistence Layer (Managed MySQL 8.0 on Railway) ]
   ├── Flyway Database Migrations (V1 to V8 Idempotent Schema & Seed Scripts)
   ├── 12 Normalized Relational Tables with Foreign Key Constraints
   └── Atomic Inventory Allocation (`daily_inventory`, `booking_allocations`)
```

Why These Technologies Were Chosen

1. **Java 21 LTS & Spring Boot 3.3.5**:

- **Concurrency & ACID Transactions**: Hospitality booking requires strict database isolation. Spring's declarative transaction management combined with Java 21 virtual thread capability ensures reliable handling of concurrent checkout sessions without race conditions.

- **Type Safety**: Financial calculations (room tariffs, GST percentages, promo deductions) are strictly bounded by `BigDecimal`, preventing floating-point inaccuracies.

2. **React 18 & Vite**:

- **Sub-Second Hill-Station Load Times**: Vite bundles compiled into modern ES modules produce a production bundle under 210 KB (gzipped), ensuring instant rendering even on congested 3G/4G mountain networks.

- **Component Decoupling**: Guest checkout and staff operational dashboards remain entirely modular with zero unnecessary re-renders.

3. **Managed MySQL 8.0**:

- **Functional Unique Indexes**: Supports ANSI SQL `CHECK` constraints and functional expressions (`COALESCE(category_id, 0)`), guaranteeing resort-wide and category-specific blackout date integrity.

- **Pessimistic Locking**: Native support for `SELECT ... FOR UPDATE` ensures no two guests can hold the same physical allotment simultaneously.

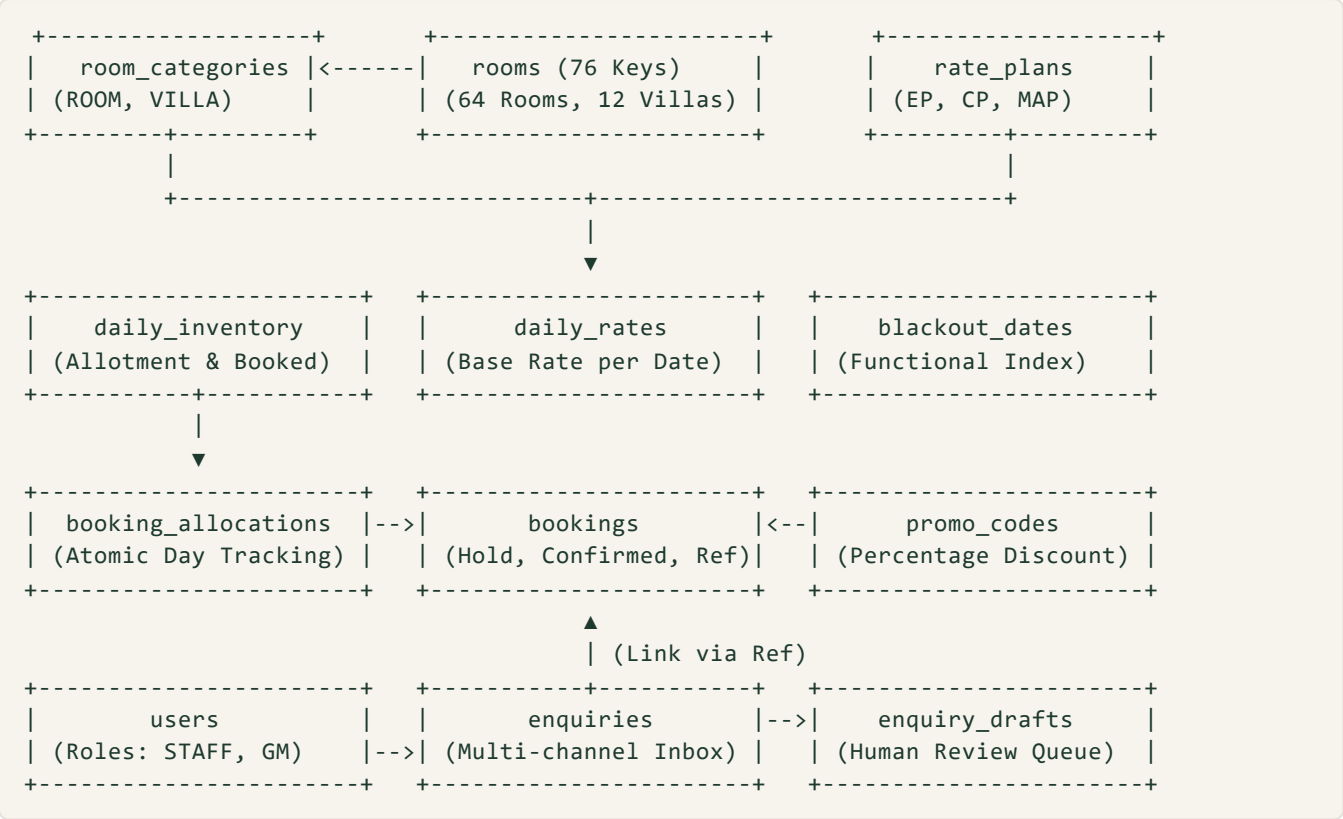
4. **Google Gemini 1.5**:

- **Structured JSON Mode**: Extracts complex multi-parameter event payloads (guest counts, date ranges, budgets) directly into strongly typed DTOs with built-in confidence scoring.

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4. Relational Data Model & Database Schema

The database schema is partitioned into 12 normalized tables managed via Flyway migrations:



Key Entity Specifications

1. **`room_categories`**: Represents the two physical accommodation types (`ROOM` with 64 units, `VILLA` with 12 units). Defines base capacities, square footage, and photo manifests.

2. **`daily_inventory`**: Tracks the admin-set website allotment (`website_allotment_total`) and active confirmed/held reservations (`booked_count`) for each date and category.

3. **`bookings`**: Stores reservation state (`HELD`, `CONFIRMED`, `CANCELLED`, `EXPIRED`), guest contact details, total price, applied promo code, and explicit UTC expiration instant (`expires_at`).

4. **`booking_allocations`**: Child table storing the discrete daily room count deductions for each reservation date.

5. **`enquiries`** & **`enquiry_drafts`**: Stores incoming customer messages, AI classification categories (`ROOM_BOOKING`, `WEDDING`, `CORPORATE_OFFSITE`, `COMPLAINT`, `UNSORTED`), confidence scores, assigned staff member, and generated response drafts pending review.

6. **`blackout_dates`**: Enforces inventory closure dates with a functional unique index:

`` sql

CREATE UNIQUE INDEX uk_blackout_dates_scope_date

ON blackout_dates (stay_date, (COALESCE(room_category_id, 0)));

``

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5. Core System Workflows & Business Logic

5.1 Three-Stage Guest Booking Funnel (Zero Overbooking Guarantee)

The guest booking journey is strictly constrained to three clear, purposeful steps:

```
[ Step 1: Details & Consent ] ──▶ [ Step 2: Payment Simulation ] ──▶ [ Step 3: Instant Confir
       (/checkout)                       (/payment/:ref)                    (/bookings/:ref)
  - Guest Name, Email, Phone        - Authoritative UTC Expiry Clock    - Reference Code (BKG-
  - DPDP 2023 Consent Checkbox      - 10-Minute Pessimistic Hold        - Detailed Price & GST
  - Promo Code Validation           - Idempotent Test Card Mock         - Direct Booking Confi
```

1. **Step 1: Guest Details & Consent (`/checkout`)**:

- The guest selects their preferred room and rate plan (`EP` Room Only, `CP` Breakfast Included, `MAP` Breakfast + Dinner).

- Validates guest full name, email, and mobile number.

- Requires explicit consent under India's Digital Personal Data Protection (DPDP) Act 2023.

- Upon form submission, the backend executes an atomic transaction in `BookingTransactionService`:

- Checks resort and category blackout dates.

- Performs `SELECT ... FOR UPDATE` row locks across each requested stay date in `daily_inventory`.

- Verifies `booked_count + requested_rooms <= website_allotment_total`.

- Increments `booked_count` and generates a booking record with `status = 'HELD'` and `expires_at = Instant.now().plus(10, ChronoUnit.MINUTES)`.

2. **Step 2: Payment Simulation (`/payment/:ref`)**:

- Displays live reservation summary and active hold countdown timer.

- **Timezone-Safe Architecture**: The backend serializes `expires_at` as an explicit UTC ISO-8601 string ending in `Z` (e.g. `2026-11-08T04:15:30Z`), preventing browser timezone shifts (such as India Standard Time UTC+5:30) from causing premature client expiration.

- The client timer serves purely as visual feedback; the backend remains the authoritative validator. Submitting payment verifies hold validity before confirming.

- Payment is processed idempotently using a UUID request key.

3. **Step 3: Instant Confirmation (`/bookings/:ref`)**:

- Updates booking status to `CONFIRMED`.

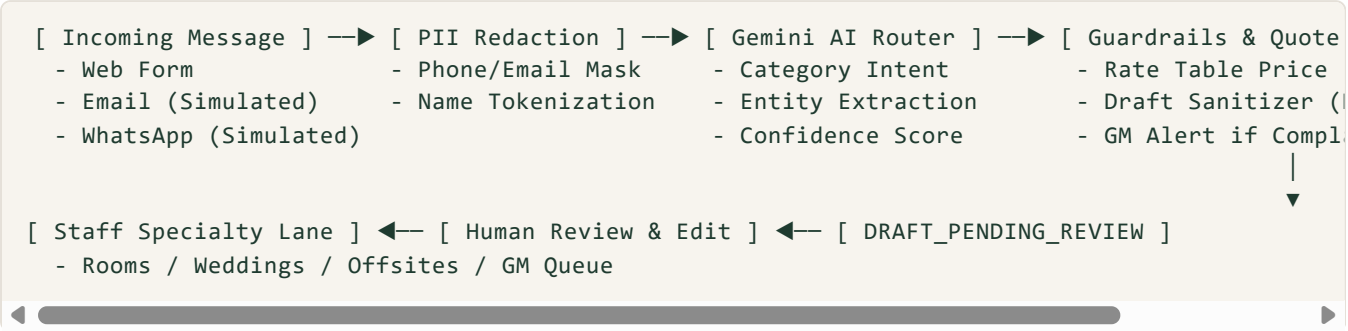
- Generates a human-readable booking reference (`BKG-XXXXXX`).

- Displays itemized room rate breakdown, promotional discount, and provisional GST.

- Does not expose public search lookups; booking access is private via confirmation link.

5.2 Multi-Channel Enquiry Intake & AI Triage Pipeline

Enquiries from the public web form (`/enquire`), forwarded emails, and WhatsApp arrive in a unified triage queue:



1. **PII Masking & Safe Prompting**:

- Senders' phone numbers and emails are masked prior to external API dispatch to comply with DPDP principles.

2. **Gemini 1.5 Classification & Extraction**:

- Categorizes enquiry into `ROOM_BOOKING` , `WEDDING` , `CORPORATE_OFFSITE` , `COMPLAINT` , or `UNSORTED` .

- Extracts structured parameters: `checkInDate` , `checkOutDate` , `guestCount` , `roomCount` , `approximateBudget` .

- Generates a contextual reply draft and computes a classification confidence score between `0.00` and `1.00` .

3. **Strict Hallucination Guardrails**:

- **Quote Injection (`QuoteInjectorService`)**: The AI is prohibited from inventing tariffs. If dates and room categories are identified, real-time rates are looked up from `daily_rates` and injected into the draft template.

- **Draft Price Sanitizer (`DraftPriceSanitizer`)**: Strips any unauthorized promotional promises or unsupported facility claims (e.g., promises of unverified AV gear or automatic room holds).

- **Safety Rules**:

- Enquiries with confidence score `< 0.70` are routed to the **Unsorted Lane** monitored by the General Manager.

- Enquiries classified as `COMPLAINT` immediately trigger a `gm_alerted = true` flag.

- **Drafts are NEVER sent automatically**: Every draft defaults to `DRAFT_PENDING_REVIEW` and requires reservations staff or GM review, editing, and one-click dispatch approval.

- Staff can re-route any misclassified enquiry with a single click.

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6. UI & Design Decisions: The 7 Core Screens

The design language embodies the client brief's requirement: "*The client values good, minimal design: clean layouts, white space, a restrained palette and clear type, with nothing on screen that lacks a purpose.*"

The color palette reflects the plantation landscape: **Deep Forest Green** (`#1E3A2F`), **Warm Sand Cream** (`#F7F4EE`), **Coffee Muted Slate** (`#5C5852`), and **Border Cream** (`#E5E0D8`).

Screen 1: The Landing Page (`/`)

- **Design Intent**: Sells the estate retreat experience without overwhelming the guest with competing actions.
- **Deduplicated Visual Hierarchy**:

  - The hero section presents high-resolution WebP photography with warm typography.

  - The floating search bar is the **single primary call-to-action** on screen, labeled `Search Availability`.

  - The top navbar contains the brand and section links, with an accessibility shortcut that smooth-scrolls to the search bar.

  - The hero text overlay removes redundant booking buttons, retaining only a secondary `Plan an Event` link for wedding and corporate planners.

Screen 2: Stay / Availability & Rooms (`/stay`)

- **Design Intent**: Instant clarity on room options and rate plans with zero friction.
- **Single Compact Filter Bar**: Features a single horizontal bar (`Check-in | Check-out | Guests | Modify Search`).
- **Auto-Fetch on Load**: Defaults to standard dates (`2026-11-14` to `2026-11-16`) or URL parameters, auto-fetching rooms immediately so guests never face an empty screen or an unnecessary initial search click.
- **Navbar Deduplication**: The navbar on `/stay` omits the generic "Check Availability" button, showing only a clean `← Back to Estate` utility link.
- **Robust Empty State**: If a date is fully booked, displays an informative card with formatted dates (never broken `"for to is"`), with actions to adjust dates or submit a direct enquiry.

Screen 3: Room Details (`/rooms/:code`)

- **Design Intent**: Immersive room exploration with immediate booking capability.
- **Preserved Stay Criteria**: Displays the selected stay dates, night count, and guest count in a clean summary strip with a prominent `Change dates` link that returns to `/stay` with all criteria preserved.
- **Dedicated Primary CTA**: Rate plan cards display clear inclusions (`EP`, `CP`, `MAP`) and a single primary button: `Book Now`.

Screen 4: Three-Step Checkout (`/checkout` ➔ `/payment/:ref` ➔ `/bookings/:ref`)

- **Design Intent**: Distraction-free, linear conversion funnel.
- **Layout**: Two-column desktop layout (guest form on the left, sticky booking summary card on the right).
- **Primary Actions**:

  - Step 1 (`/checkout`): Single primary button `Proceed to Payment`.

  - Step 2 (`/payment/:ref`): Clear 10-minute countdown indicator and primary button `Simulate & Confirm Reservation`.

  - Step 3 (`/bookings/:ref`): Complete confirmation breakdown with reference code and printable receipt.

Screen 5: Staff Enquiry Inbox (`/staff/enquiries`)

- **Design Intent**: High-velocity operational triage hub for the reservations team and GM.
- **Layout**:

  - Left panel: Filterable specialty tabs (`All` , `Rooms` , `Weddings` , `Offsites` , `Complaints` , `Unsorted`).

  - Enquiry cards display channel icon (Web, Email, WhatsApp), sender, confidence score badge, and GM alert indicator.

  - Right detail panel: Displays the raw customer message, AI extracted parameters, 1-click re-route dropdown, editable draft response, and one-click dispatch button.

Screen 6: Staff Rate & Allotment Admin (`/staff/inventory`)

- **Design Intent**: Comprehensive inventory management preventing overbooking.
- **Capabilities**:

  - Interactive calendar grid allowing staff to view and modify daily base rates, total website allotment, and booked counts across categories.

  - Blackout date management for resort-wide or category-specific closures.

  - Promotional discount code manager (active dates, discount percentages, usage tracking).

  - One-click **Daily Booking CSV Export** sanitized against spreadsheet injection (CWE-1236).

Screen 7: Structured Event & Wedding Form (`/enquire`)

- **Design Intent**: Guided multi-parameter intake for complex destination events.
- **Capabilities**:

  - Interactive category selector (`Weddings` , `Corporate Offsites` , `Individual Stays`).

  - Contextual fields for estimated guest counts, preferred dates, room blocks, catering preferences, budget estimates, and conference setup needs.

  - Neutral default choices (e.g. conference hall selection defaults to "Not specified / please advise").

  - Single primary CTA: `Submit Enquiry` .

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7. Scope Cuts & Future Roadmap

In strict accordance with the client brief's *CORE BUILD* guidelines and *OUT OF SCOPE* boundaries:

What Was Cut (and Why)

1. **Channel Manager / OTA Two-Way Sync**: Explicitly declared out of scope by the brief. The resort controls website overbooking by allocating an admin-set website allotment (`website_allotment_total`) that acts as an isolated inventory pool.

2. **Production Payment Gateway**: Replaced with an internal deterministic test payment simulator (`POST /api/v1/public/payments/{ref}/mock-webhook`), eliminating non-commercial financial transaction compliance while validating the complete state machine.

3. **Loyalty Program & Restaurant Reservations**: Excluded to focus development velocity exclusively on the core direct-booking engine and AI triage workflows.

4. **Outbound Third-Party Telephony/SMS Rails**: Replaced with internal simulated delivery adapters (`EMAIL_SIM`, `WHATSAPP_SIM`) generating verified delivery IDs.

What to Do Next (Production Roadmap)

1. **Production Payment Gateway Integration**: Connect Razorpay / Stripe Webhooks with UPI intent flow for Indian domestic travelers.

2. **PMS Webhook Integration**: Implement bidirectional synchronization with hotel property management systems (e.g. Opera, Cloudbeds) for automated daily reservation synchronization.

3. **WhatsApp Business Cloud API**: Transition simulated WhatsApp ingestion to official Meta Cloud API webhooks with end-to-end template verification.

4. **GM Financial Dashboard**: Implement a real-time analytics view tracking saved OTA commission metrics (moving towards the target ₹33.3 lakh/year saving).

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8. Responsive Viewport Compliance Matrix

All guest and staff interfaces were validated via headless Chrome across target responsive breakpoints:

Viewport Breakpoint	Dimensions	Tested Routes	Horizontal Overflow	Result
Mobile (iPhone 14 / Compact)	390px × 844px	`/`, `/stay`, `/rooms/:code`, `/checkout`, `/enquire`, `/staff/enquiries`	**Zero** (`scrollWidth = 390px`)	**PASS**
Tablet (iPad Portrait)	768px × 1024px	`/`, `/stay`, `/checkout`, `/staff/inventory`	**Zero** (`scrollWidth = 768px`)	**PASS**
Desktop (Standard Laptop)	1280px × 800px	All guest routes & staff dashboards	**Zero** (`scrollWidth = 1280px`)	**PASS**

Tap targets across all interactive mobile elements meet or exceed the **44px × 44px** standard (`var(--tap-min, 2.75rem)`).